



Application of Electricity on the body

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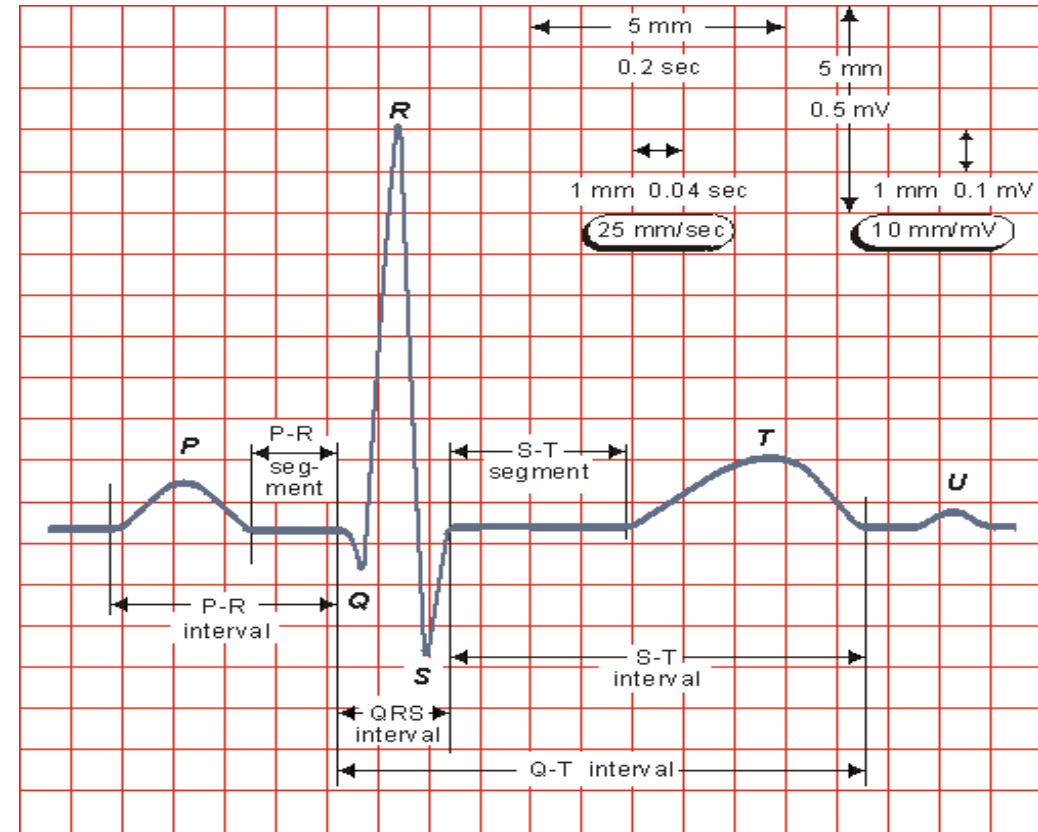
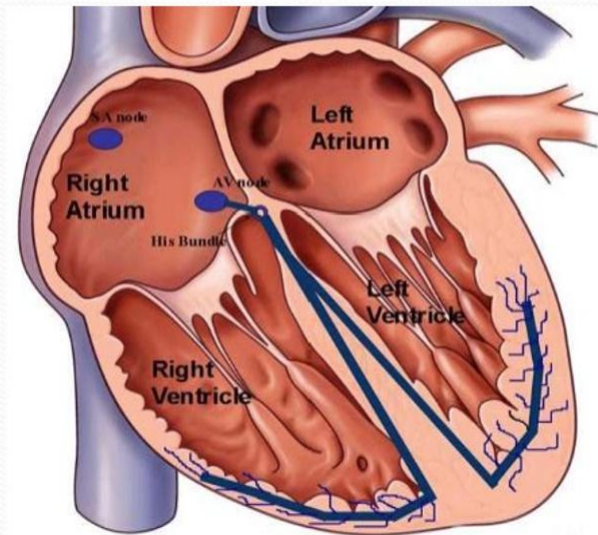
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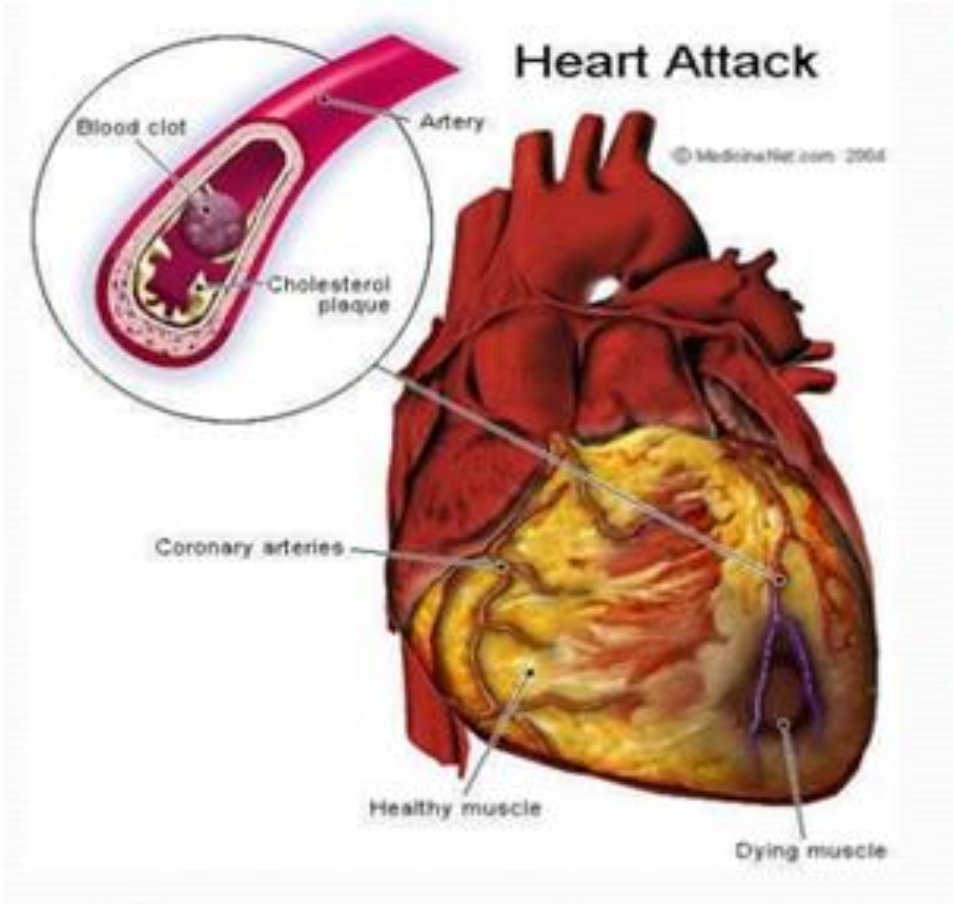
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1. DEFIBRILATOR

Electrical pattern





- Occlusion of the coronary artery led to ischemia
- Ischemia leads to infarct which causes interruption of normal cardiac conduction
- Infarct = VF/VT

Symptoms of VF include:

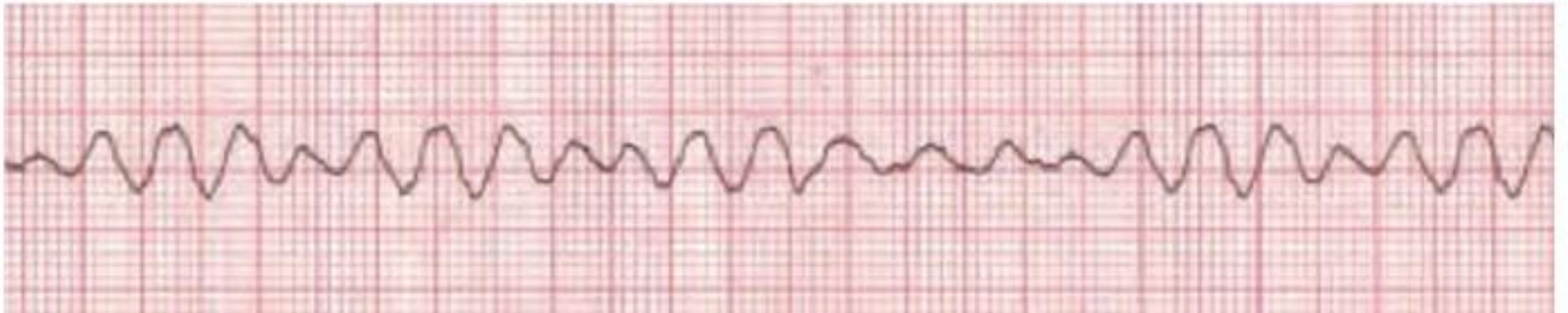
- Near fainting or transient dizziness
- Fainting
- Acute shortness of breath
- Cardiac arrest

ECG Findings

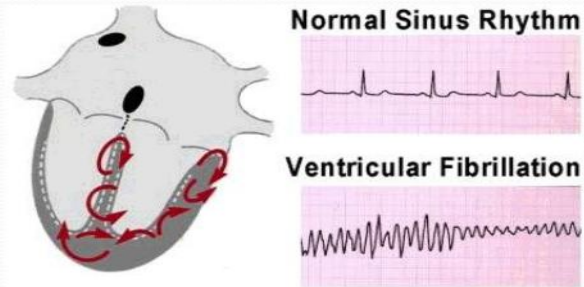
Fibrillation waves of varying amplitude and shape.

No identifiable P waves, QRS complexes, or T waves

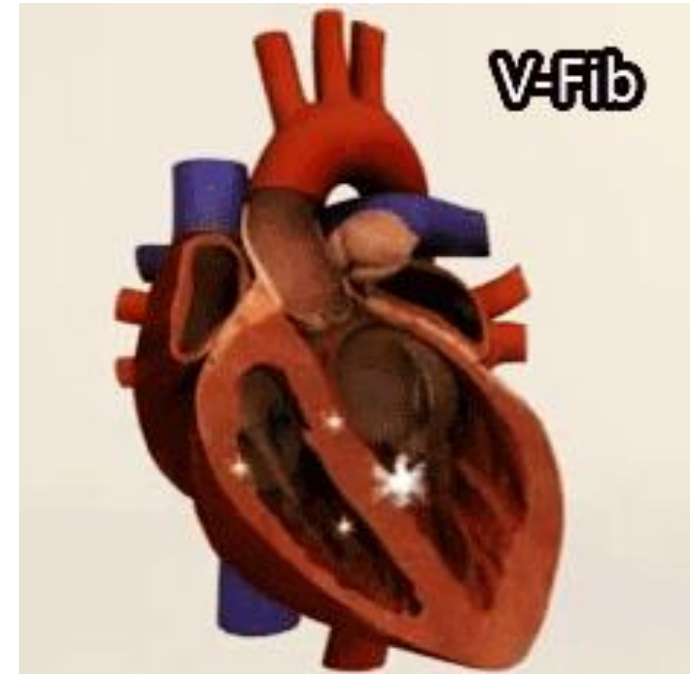
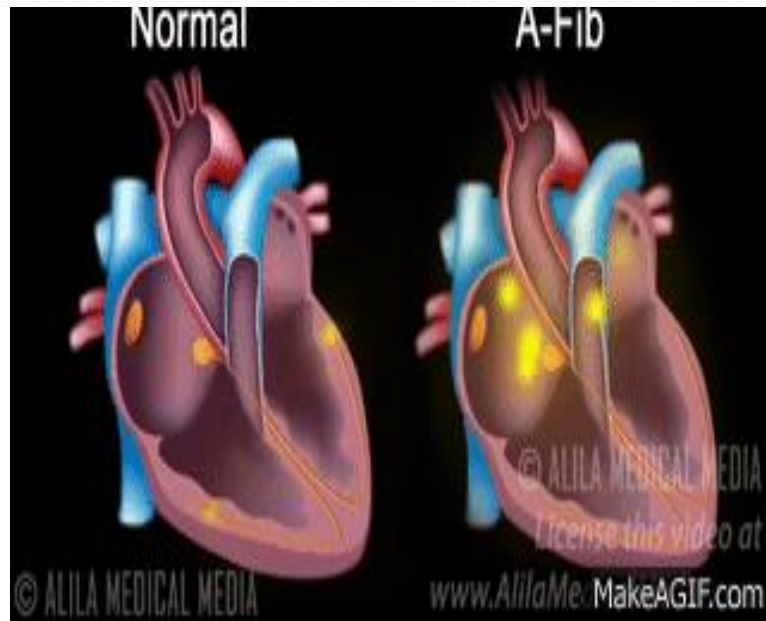
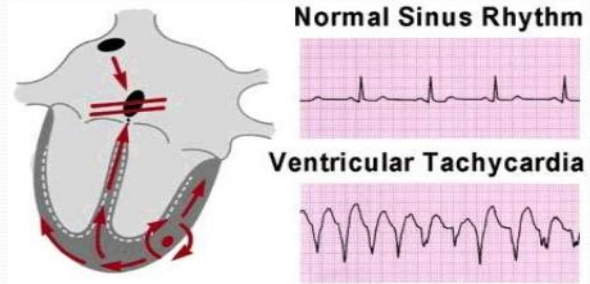
Heart rate anywhere between 150 to 500 beat/min



Ventricular Fibrillation



Ventricular Tachycardia



Ventricular Fibrillation

- Ventricular fibrillation is a serious cardiac emergency resulting from asynchronous contraction of the heart muscles
- Due to ventricular fibrillation, there is an irregular rapid heart rhythm.
 - cause the heart to stop pumping blood
 - leading to brain damage

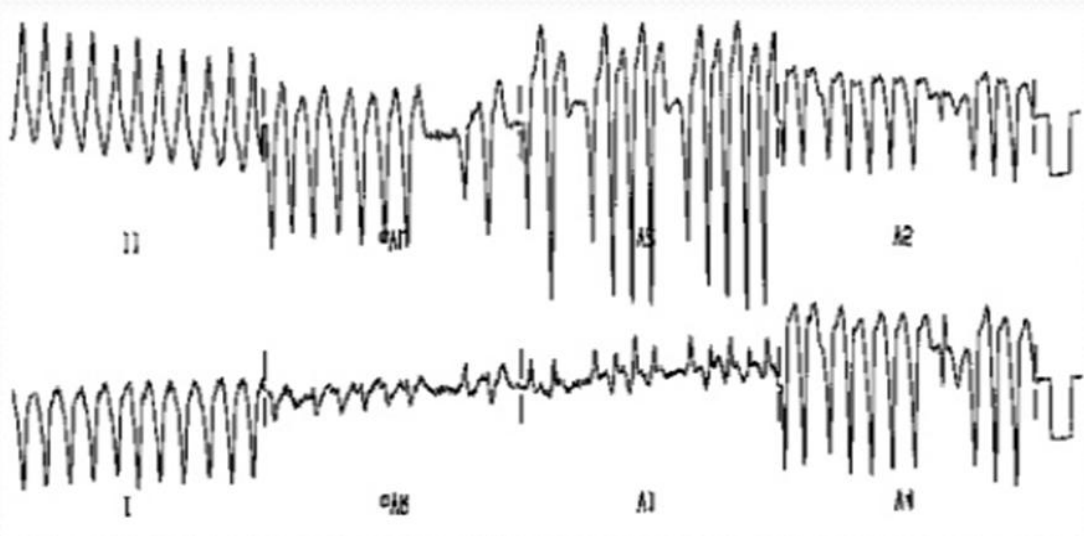


Fig. Ventricular fibrillation



Fig. Normal heart beat

Ventricular fibrillation

- **Causes**

- Drugs
 - Electrolytic disturbances
- Coronary thrombosis
 - Atherosclerotic coronary artery disease
 - Ischemia and infarction of cardiac tissue
- Accidental electrocution
 - The passage of a small electric current
 - (30 μ A through human heart)

The Treatment

- Depolarization all the cardiac muscle simultaneously, as the pacemaker (SA) have a higher intrinsic discharge rate than the rest of the cardiac muscle, they should be able to resume their normal pacing function.

The Treatment

1. Cardiopulmonary resuscitation (CPR) to keep blood moving

https://www.physio-pedia.com/images/d/df/Chest_compressions.gif

2. Defibrillation/Cardioversion

Defibrillator

A defibrillator is a medical device that delivers a controlled electrical shock to the heart to terminate life-threatening arrhythmias.

What is the goal of defibrillation?

To restore normal sinus rhythm by simultaneous depolarization of myocardial cells.

- ☐ Ventricular fibrillation and pulseless ventricular tachycardia.

What happens to cardiac cells during defibrillation?

All myocardial cells are depolarized at once, allowing the SA node to regain control.

Types of Defibrillators

1. External Defibrillators:

- Manual External Defibrillators (used by professionals).
- Automated External Defibrillators (AEDs, for public use).

2. Implantable Cardioverter Defibrillators (ICDs):

- Surgically implanted for patients at risk of sudden cardiac arrest.

3. Wearable Defibrillators:

Temporary external devices for high-risk patients

External



Physical Principles of Defibrillation

- **Electric Shock Delivery:**

- A controlled electric current is passed through the heart.
- Energy range: Typically 120–360 Joules for external defibrillators.

- **Waveforms:**

- Biphasic waveform (modern defibrillators): More efficient, requires less energy.
- Monophasic waveform (older models): Higher energy required.

- **Current Flow:**

- Current passes between two electrodes (paddles or pads) placed on the patient's chest.

Key Components of a Defibrillator

1. Power Source:

- Battery (rechargeable or disposable) for portability.

2. Capacitor:

- Stores and delivers the electrical energy.

3. Electrodes/Paddles:

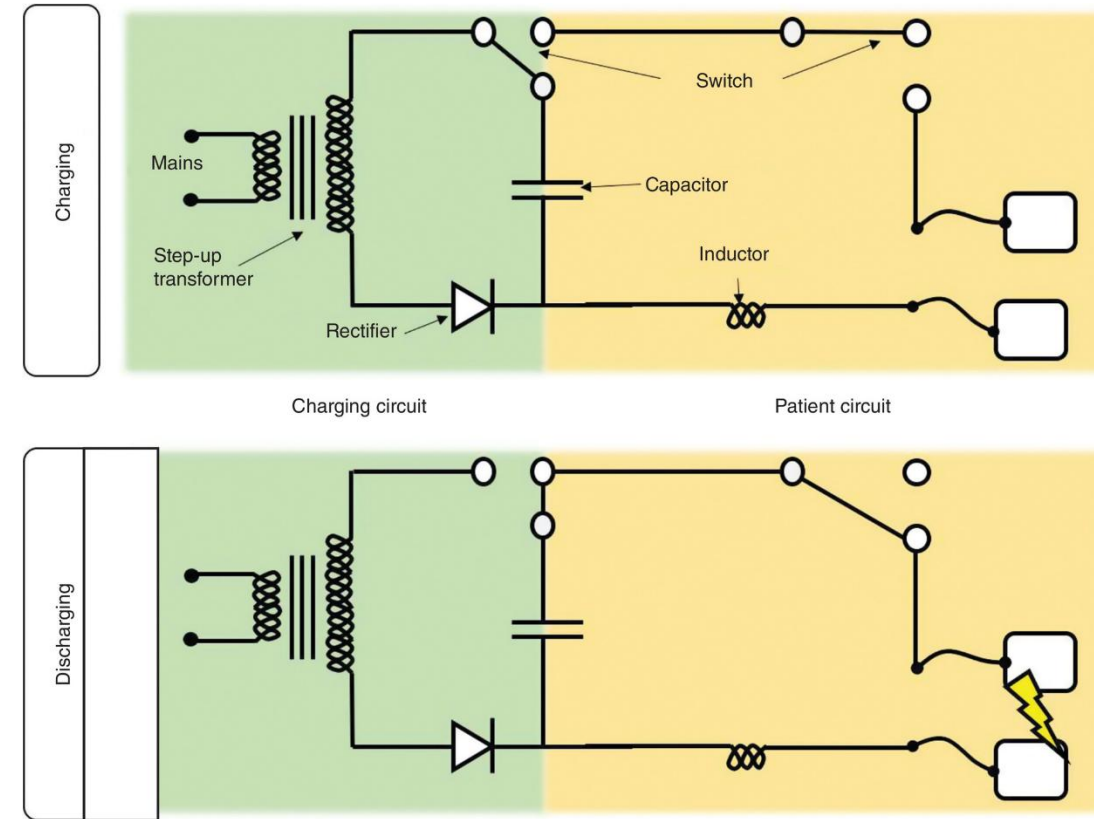
- Conduct electricity to the patient's chest.

4. Control Unit:

- Microprocessor for energy selection, timing, and safety checks.

5. ECG Monitor:

- Displays the patient's heart rhythm.



Energy Delivery and Dosage

- **Energy Levels:**

- Initial shock: 120–200 Joules (biphasic) or 360 Joules (monophasic).
- Subsequent shocks: Adjusted based on patient response.

- **Factors Affecting Energy Delivery:**

- Patient impedance (resistance of the chest wall).
- Electrode placement and size.
- Type of waveform used.

Electrical cardioversion (Defibrillator) –

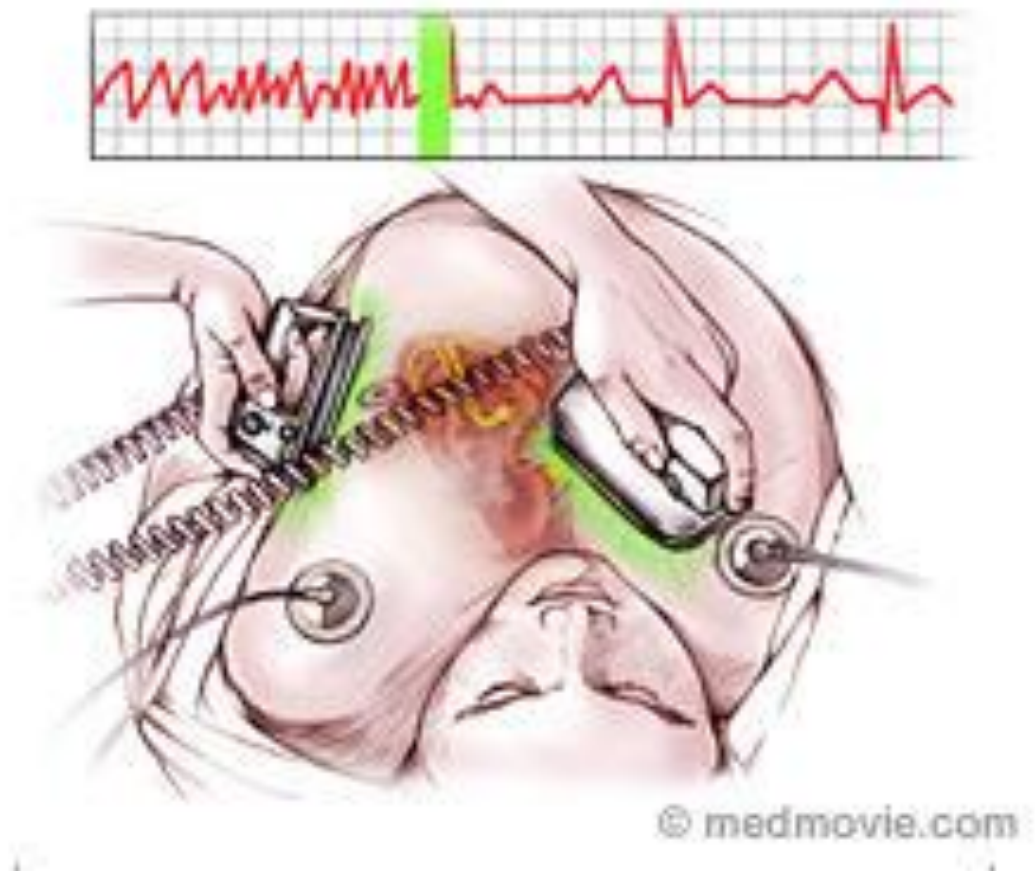
- ☐ Instantaneous mode (immediately applied)
- ☐ Synchronized mode.

(The discharge pulse is not immediately applied to the subject. The signal delayed to occur during the falling part of the R-wave.)

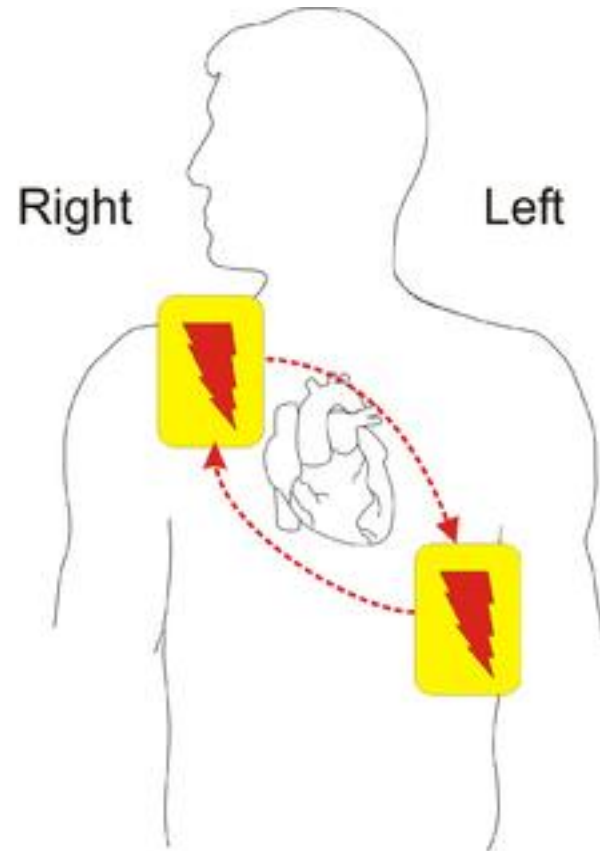
Q: A patient's heart has not stopped beating but is beating erratically. Which defibrillator mode is used to restore a normal rhythm?

- A. Asynchronous defibrillation
- B. Synchronized cardioversion
- C. External pacing
- D. Vagal maneuver

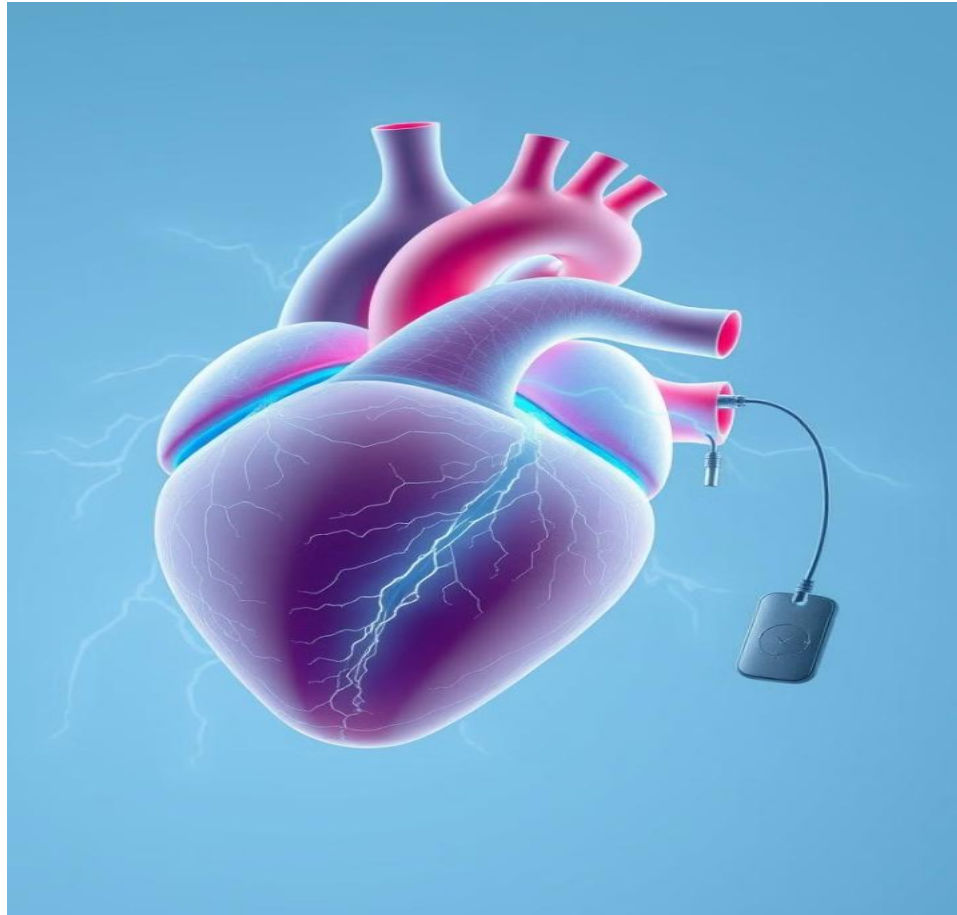
Feature	Defibrillation	Synchronized Cardioversion
Indication (core)	Cardiac arrest with chaotic rhythm	Unstable patient with organized rhythm
Rhythm type	Disorganized	Organized but abnormal
Common rhythms treated	VF, pulseless VT	AF, atrial flutter, SVT, VT with pulse
Pulse present? 🔔	NO	YES
Synchronization	✗ Not synchronized	☑ Synchronized with R wave
ECG requirement	ECG not needed	Clear QRS required
Energy timing	Delivered immediately	Delivered at R wave
Risk avoided	—	R-on-T → ventricular fibrillation
Sedation required?	✗ No (patient unconscious)	☑ Yes (if conscious)
Energy level	Higher	Lower
Emergency level	Immediate life-saving	Urgent but controlled



The important thing is getting the current to flow through the heart, so **where the paddles are applied is crucial**. One way of applying them is **to put one paddle above and to the right of the heart and the other slightly beneath and to the left**



2. Artificial Pacemaker



A 72-year-old man presents with repeated episodes of dizziness and fainting over the past month. He complains of fatigue and reduced exercise tolerance. On examination, his pulse rate is **38 beats/min** and regular. Blood pressure is 90/60 mmHg. ECG shows **complete heart block** with atrioventricular dissociation and a slow ventricular escape rhythm.

Most appropriate management:

- ☐ **Permanent pacemaker implantation**

Pacemaker

A pacemaker is a small, battery-operated, implantable medical device that monitors and regulates heart rhythm by delivering low-energy electrical impulses to the heart muscle. Used primarily to treat arrhythmias (such as slow, fast, or irregular heartbeats) or heart failure, it helps ensure the heart beats at a normal rate.

Components

- A pacemaker consists of
- a pulse generator (The pulse generator houses the circuitry that controls pacing.
 - a battery, and
 - leads that deliver electrical impulses to the heart.

Functions

Pacemakers monitor heart rhythm, detect slow heart rates, and deliver electrical pulses to regulate heartbeats. This ensures a steady and regular heartbeat.

Pacemaker Implantation Procedure

1

Preparation

The patient undergoes local anesthesia, and the procedure is performed in a sterile environment.

2

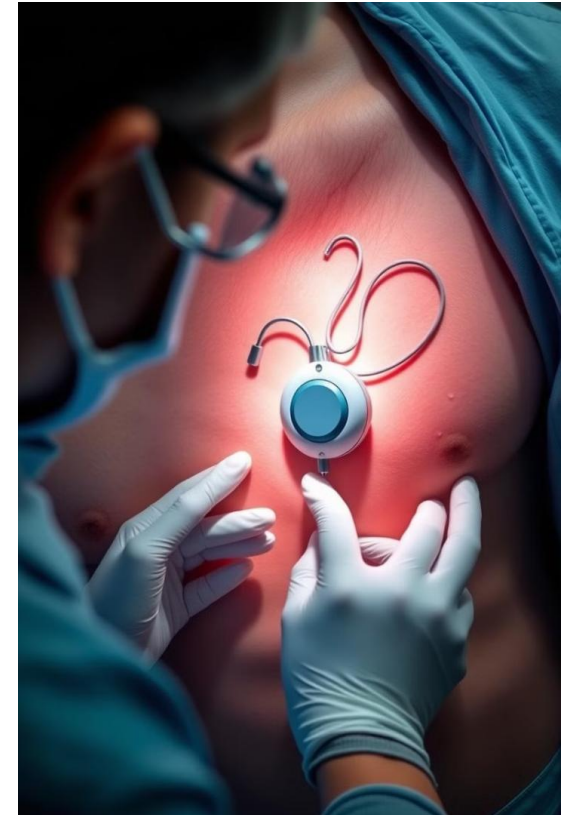
Insertion

The pacemaker is inserted under the skin, typically in the chest area, and leads are connected to the heart.

3

Monitoring

Post-implantation, the pacemaker is monitored to ensure proper functioning and adjustments are made if needed.





Heart's Electrical System and Pacemaker Function

1

Natural Rhythm

The heart's natural pacemaker, the sinoatrial (SA) node, generates electrical signals that control the rhythm of the heart.

2

Pacemaker Role

In cases of slow heartbeat or heart block, the pacemaker takes over, stimulating the heart to beat at a predetermined rate.

3

Regulation

The pacemaker senses the heart's rhythm and only delivers pulses when necessary, adjusting the pacing rate to maintain a healthy heartbeat.

Pacemaker Sensing and Stimulation Mechanisms

Sensing

- 1 The pacemaker constantly monitors heart electrical activity to detect slow heart rates or abnormal rhythms.
-

Stimulation

- 2 When the pacemaker detects a slow heart rate, it delivers a small electrical pulse to the heart muscle, stimulating it to beat faster.
-

Regulation

- 3 This process ensures a consistent and regular heartbeat, providing the heart with the necessary electrical impulses to function properly.

Pacemaker Power Source and Longevity

1

Battery

2

Lifespan

Pacemaker batteries typically last for 5-10 years, depending on usage and battery type.

3

Replacement

When the battery nears its end, a simple outpatient procedure replaces the battery without replacing the entire device.

Overview of Pacemaker Types

Single-Chamber

- Stimulates only one heart chamber (atria or ventricles)
VVI, AAI

Dual-Chamber

- Stimulates both the atria and ventricles, coordinating heart rhythm
DDD

A biventricular :

- Three-lead pacemaker—often part of

Cardiac Resynchronization Therapy (CRT) designed to improve the heart's pumping efficiency in patients with advanced heart failure.

Programmable Pacemakers

- A **programmable pacemaker** is an implantable cardiac device that allows healthcare providers to non-invasively adjust its settings (e.g., pacing mode, rate, sensitivity, and output) to meet the specific needs of the patient. This flexibility ensures optimal pacing therapy and adaptability to changing clinical conditions over time.

A patient with complete heart block but preserved sinus node activity requires pacing that preserves atrial contribution to ventricular filling. Which is the **best pacing option**?

- A. VVI
- B. AAI
- C. DDD
- D. VOO

Correct Answer: C

Benefits of Dual-Chamber Pacemakers

1 Improved Heart Function

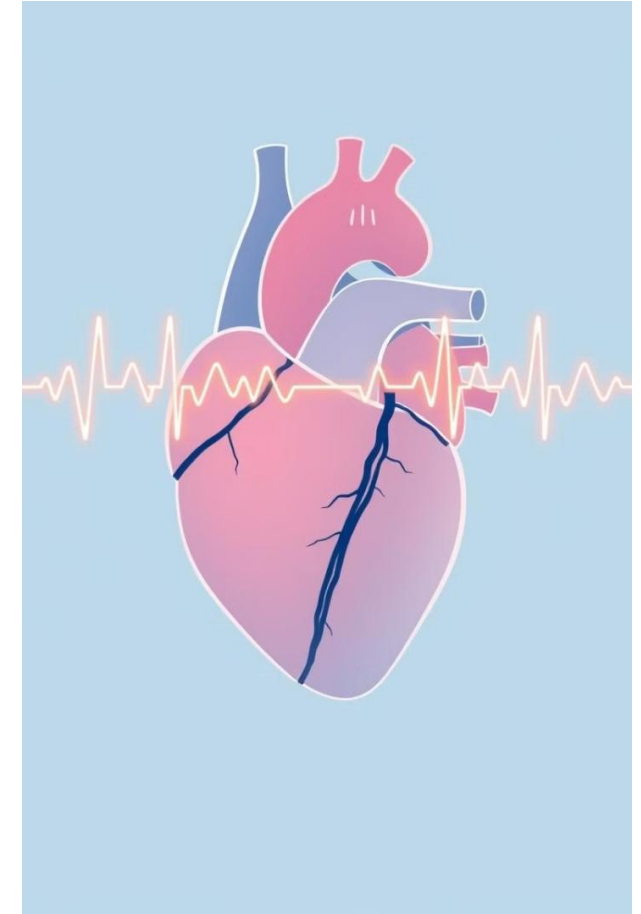
Optimal synchronization enhances blood flow and heart performance

2 Reduced Symptoms

Alleviates symptoms like fatigue, dizziness, and shortness of breath

3 Enhanced Quality of Life

Patients experience improved physical capacity and overall well-being



Programmable Features of Modern Pacemakers



Rate Adjustment

Pacemaker can adjust heart rate based on physical activity



Sensitivity Control

Pacemaker responds to changes in heart rhythm and adjusts accordingly



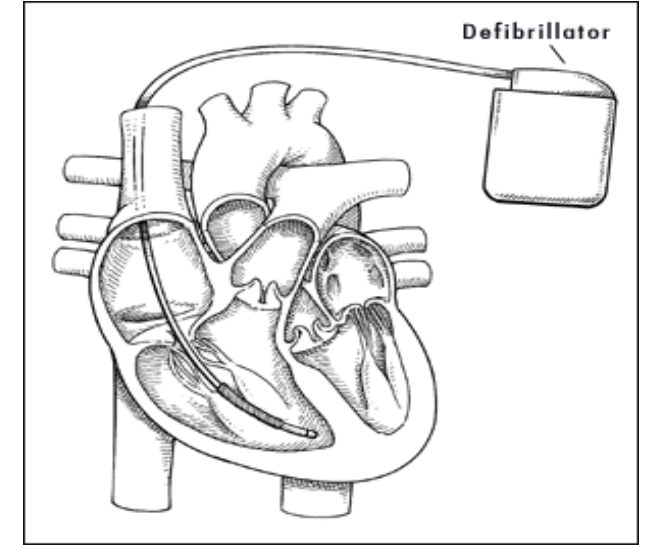
Telemetry Monitoring

Wireless transmission of data to healthcare providers for remote monitoring



The ICD constantly monitors heart rhythm.
Upon detection of VT/VF delivers a programmed treatment.
Capable of applying variety of possible treatments:

- Pacing - deliver a sequence of low-amplitude pulses.
 - Cardioversion - a mild shock (if pacing fails in terminating VT).
 - Defibrillation - a large shock to terminate VF.
 - Pacemaker - for slow heartbeat, can act as pacemaker.



Advancements and Future Developments in Pacemaker Technology



Miniaturization

Pacemakers are becoming smaller and more discreet, with advancements in materials and circuitry.



Wireless Monitoring

Wireless technology enables remote monitoring and adjustments, enhancing patient care and reducing clinic visits.



Smart Pacing

The development of adaptive pacing systems, adjusting to changing needs and improving responsiveness.



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